

Quiz 9: Plastics

Review of Session 9. ~15 minutes.

Section A: Density

1. Which three plastics float in water?
2. A sample sinks in isopropyl alcohol (0.79) but floats in vegetable oil (0.92). Which plastic?
3. A sample floats in water but sinks in vegetable oil. Which two plastics is it likely to be, and how do you tell them apart?
4. A sample sinks in water but floats in saturated salt water (~1.20). Which plastic?

Section B: Resin codes

Match the code to the plastic: 1, 2, 3, 4, 5, 6.

Section C: Types & history

1. What is the difference between a thermoplastic and a thermoset?
 2. "All plastics in the rules occur in thermo_____ form." Fill in the blank.
 3. What was the first fully synthetic plastic? Who invented it, and when?
 4. About what percentage of plastic is recycled?
 5. How do you tell LDPE from HDPE by feel?
 6. Which plastic produces dense black sooty smoke when burned?
 7. Which plastic self-extinguishes and smells acrid (like HCl) when burned?
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Answer Key

Section A

1. PP, LDPE, HDPE.
2. PP (density ~0.90–0.92).
3. LDPE and HDPE — squeeze test: soft and stretchy = LDPE; stiff and crinkly and cloudier = HDPE.
4. Polystyrene (PS) — density ~1.04–1.07, floats in salt water at 1.20.

Section B 1 = PETE, 2 = HDPE, 3 = PVC, 4 = LDPE, 5 = PP, 6 = PS.

Section C

1. Thermoplastic softens and can be re-melted and remolded repeatedly; thermoset is cross-linked and cured once — it chars instead of melting and cannot be remolded.
2. Thermoplastic.
3. Bakelite; Leo Baekeland; 1907.
4. About 9%.
5. LDPE is soft and flexible; HDPE is stiffer, cloudier, crinklier.
6. Polystyrene (PS).
7. PVC.